# V. EDA – DATA VISUALIZATION

**AIM:**

To explore and understand the underlying patterns, distributions, and relationships within the dataset through visual representations, which aids in uncovering insights, detecting anomalies, and guiding further analysis.

**PROGRAM:**

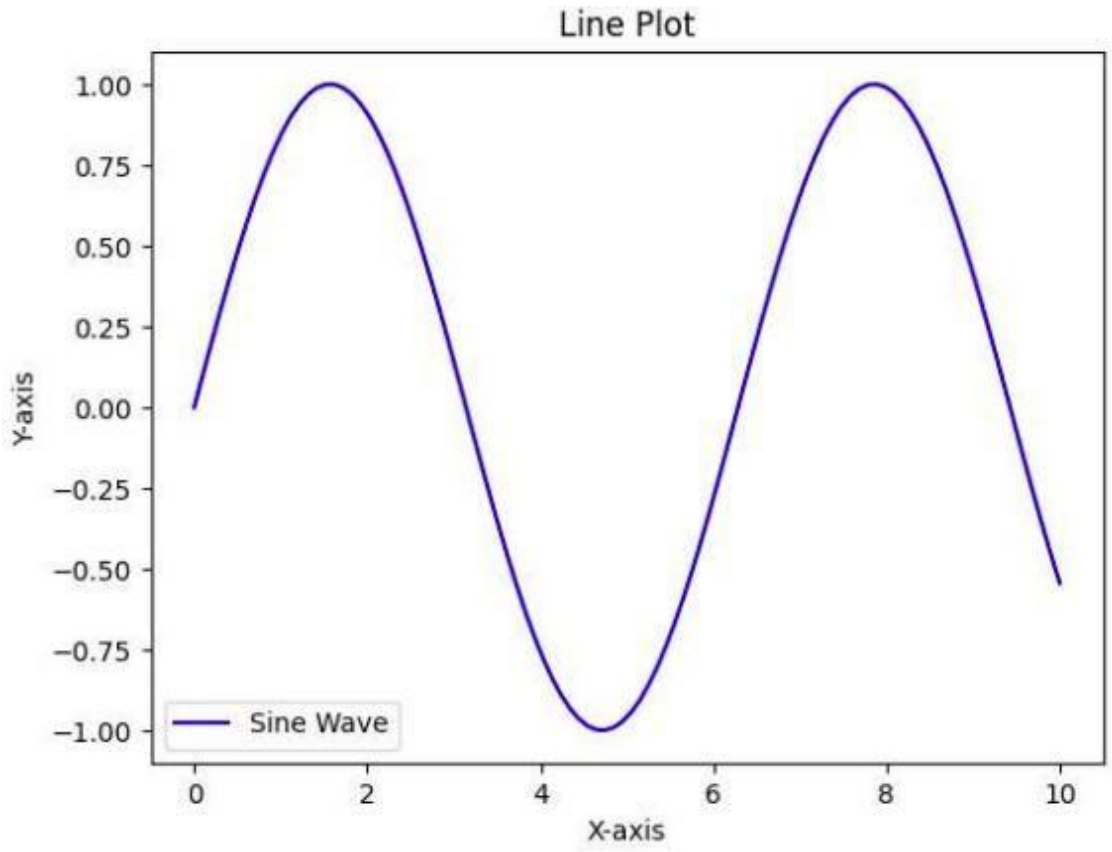
```
import matplotlib.pyplot as plt
import pandas as pd
import numpy as np

df = pd.read_csv("/content/data.csv") # Replace with your file
df.head()
```

| | Make | Model | Year | Engine Fuel Type | Engine HP | Engine Cylinders | Transmission Type | Driven_Wheels | Number of Doors | Market Category | Vehicle Size | Vehicle Style | highway MPG | city mpg | Popularity | MSRP |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | BMW | 1 Series M | 2011 | premium unleaded (required) | 335.0 | 6.0 | MANUAL | rear wheel drive | 2.0 | Factory Tuner,Luxury,High-Performance | Compact | Coupe | 26 | 19 | 3916 | 46135 |
| 1 | BMW | 1 Series | 2011 | premium unleaded (required) | 300.0 | 6.0 | MANUAL | rear wheel drive | 2.0 | Luxury,Performance | Compact | Convertible | 28 | 19 | 3916 | 40650 |
| 2 | BMW | 1 Series | 2011 | premium unleaded (required) | 300.0 | 6.0 | MANUAL | rear wheel drive | 2.0 | Luxury,High-Performance | Compact | Coupe | 28 | 20 | 3916 | 36350 |
| 3 | BMW | 1 Series | 2011 | premium unleaded (required) | 230.0 | 6.0 | MANUAL | rear wheel drive | 2.0 | Luxury,Performance | Compact | Coupe | 28 | 18 | 3916 | 29450 |
| 4 | BMW | 1 Series | 2011 | premium unleaded (required) | 230.0 | 6.0 | MANUAL | rear wheel drive | 2.0 | Luxury | Compact | Convertible | 28 | 18 | 3916 | 34500 |

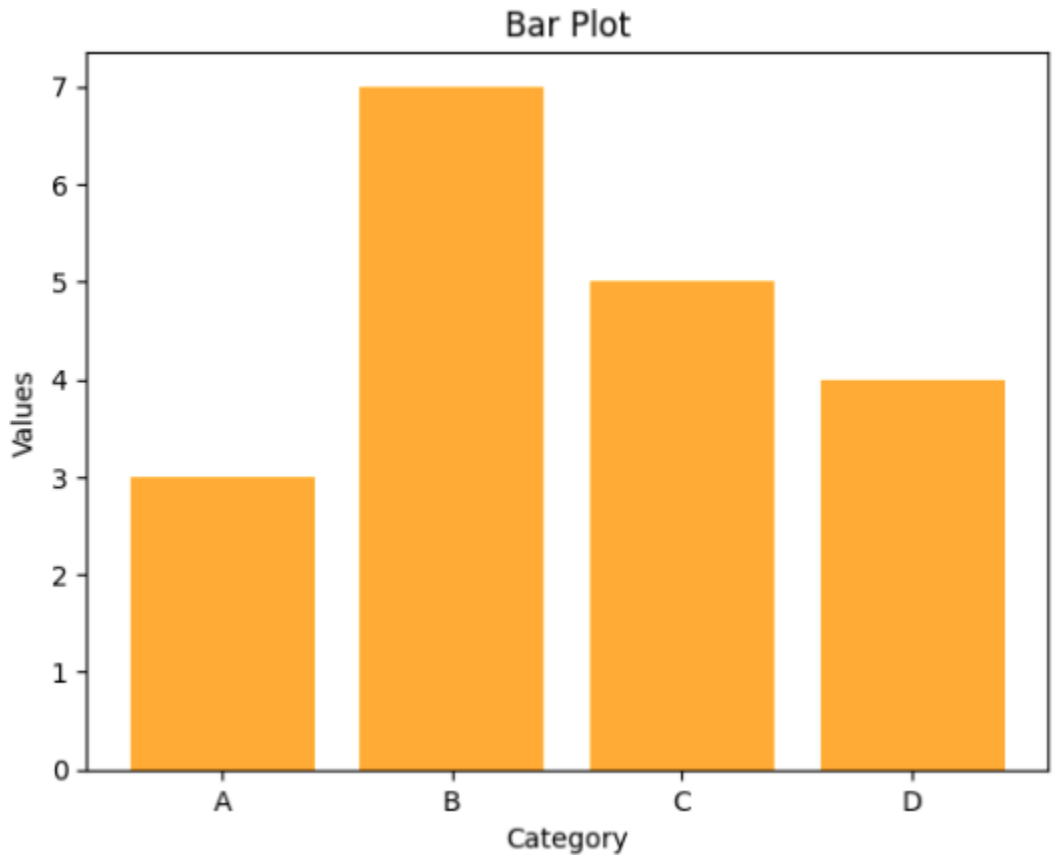
**LINE CHART:**

```
x = np.linspace(0, 10, 100)
y = np.sin(x)
plt.plot(x, y, color='blue', label='Sine Wave')
plt.xlabel('X-axis')
plt.ylabel('Y-axis')
plt.title('Line Plot')
plt.legend()
plt.show()
```



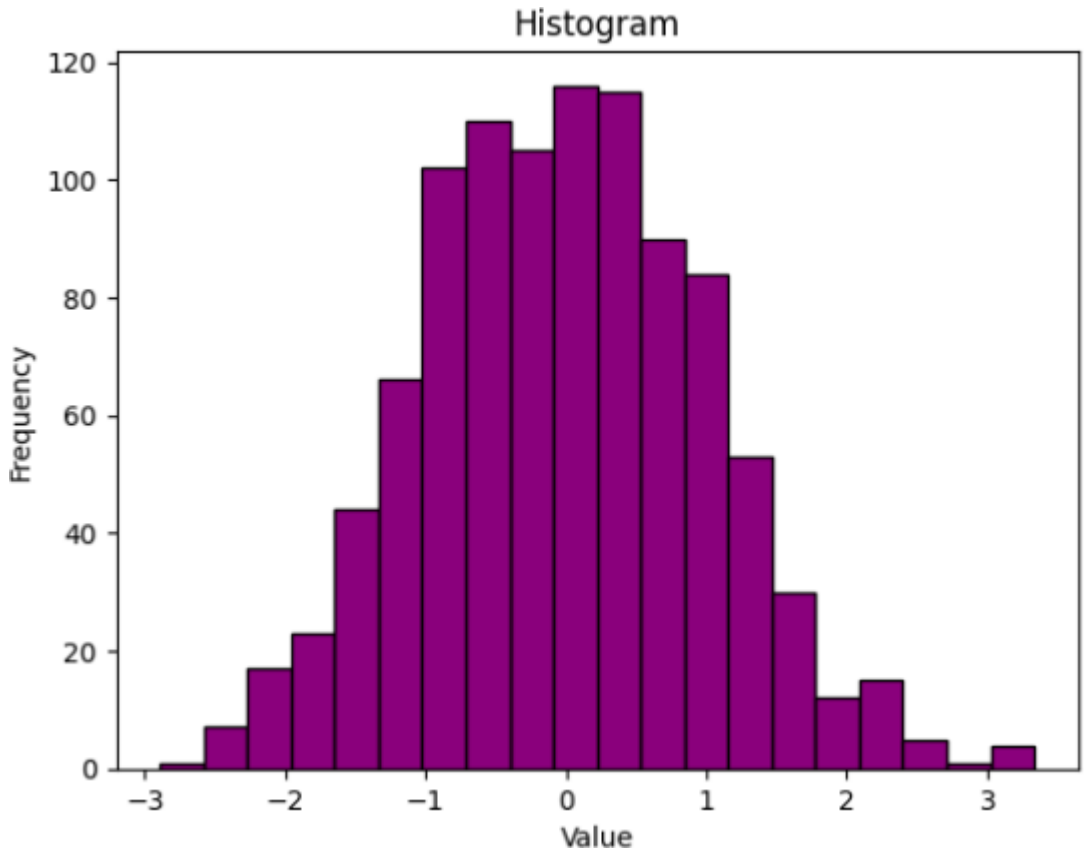
**BAR CHART:**

```
categories = ['A', 'B', 'C', 'D']
values = [3, 7, 5, 4]
plt.bar(categories, values, color='orange')
plt.xlabel('Category')
plt.ylabel('Values')
plt.title('Bar Plot')
plt.show()
```



**HISTOGRAM:**

```
data = np.random.randn(1000)
plt.hist(data, bins=20, color='purple', edgecolor='black')
plt.xlabel('Value')
plt.ylabel('Frequency')
plt.title('Histogram')
plt.show()
```




**RESULT:**

Thus, the given program was written and executed successfully.